SYSTEM EXECUTION, RUN & DEMONSTRATION GUIDE

Step-by-Step Setup Instructions, Environment Configuration & College Viva Demo Script

SMART WORKPLACE SAFETY INTELLIGENCE PLATFORM | PHASE 1 (40%) DEPLOYMENT MANUAL

1. Hardware & Software Prerequisites

Component	Minimum Requirement	Recommended Specification
Operating System	Windows 10/11 (64-bit) or Ubuntu 20.04+ LTS	Windows 11 or Ubuntu 22.04 LTS
Python Environment	Python 3.11.x (with pip)	Python 3.11.x inside dedicated venv
Node.js & NPM	Node.js v18+ & NPM v9+	Node.js v20+ or v22+ LTS
Processor (CPU)	Intel Core i5 (8th Gen) / AMD Ryzen 5	Intel Core i7 (11th+ Gen) / AMD Ryzen 7
GPU Acceleration	CPU-only mode supported (15-25 FPS)	NVIDIA GeForce RTX 3060+ with CUDA 12.x
System Memory (RAM)	8 GB RAM	16 GB DDR4/DDR5 RAM
Camera Device	Integrated standard 720p Laptop Webcam	External 1080p USB Webcam / Industrial IP Stream

2. Step-by-Step Installation & Setup

Step 1: Clone Repository & Configure Environment

Open a PowerShell terminal (or Bash on Linux) and navigate to the project root directory. Copy the environment configuration template:

```
cd c:\GUIDO\workplace-safety
copy .env.example .env # (Linux: cp .env.example .env)
```

Step 2: Python Virtual Environment & Backend Setup

Create an isolated Python 3.11 virtual environment and install all required computer vision and web server dependencies:

```
python -m venv venv
.\venv\Scripts\activate # (Linux: source venv/bin/activate)
pip install -r backend/requirements.txt
```

Step 3: Download & Verify YOLOv8n Weights

Execute the automated model initialization script to download pre-trained weights to ml/models/yolov8n.pt:

```
python scripts/download_model.py
```

Step 4: Frontend Installation

Navigate to the frontend folder and install the React and Tailwind UI component dependencies:

```
cd frontend
npm install
cd ..
```

3. Launching the Platform Services

Terminal 1 — Launch Backend FastAPI Server (Port 8000):

```
cd c:\GUIDO\workplace-safety\backend
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

• Interactive Swagger API Documentation: **http://localhost:8000/docs**
• Health Verification Endpoint: **http://localhost:8000/api/health**

Terminal 2 — Launch Frontend React Dashboard (Port 3000 / 5173):

```
cd c:\GUIDO\workplace-safety\frontend
npm run dev
```

• Open your web browser to: **http://localhost:3000** (or Vite port shown in console)

DOCKER SHORTCUT: DOCKER ONE-COMMAND ALTERNATIVE: If Docker is installed on your system, you can start both services with zero configuration: cd docker && docker compose up --build. The frontend is served at port 3000 with automatic Nginx proxying to the FastAPI backend.

4. College Project Demonstration Script (Viva Presentation)

Follow this exact 7-step presentation protocol during your academic project evaluation to deliver an impressive, rock-solid technical demonstration:

Step & Action	What to Do on Screen	What to Say & Explain to Evaluators
Step 1: Dashboard Overview & Health	Open http://localhost:3000. Show KPI cards, system status ('Live'), and event breakdown.	*'Our platform uses an enterprise Light Theme. Notice all metrics are connected to live database/WebSocket feeds with zero hardcoded values.'*
Step 2: Safety Zone Defining Walkway	Go to **Safety Zones (`/zones`)**. Click 'Create Zone', drag a rectangle across the canvas, name it 'Main Walkway', threshold 30%, persistence 10 frames.	*'We define a geometric safety zone. The system stores normalized coordinate vectors [0-1] so the zone applies accurately regardless of camera resolution.'*
Step 3: Live Detection Webcam / File Ingestion	Go to **Live Monitoring (`/monitoring`)**. Select 'Webcam' (or 'Video File'), click **Start Detection**.	*'FastAPI launches a background inference loop. Notice the real-time FPS counter (~25-35 FPS) and inference latency (~30ms) on the bottom bar.'*
Step 4: Worker Detection Bounding Boxes	Stand in front of webcam or show video footage. Point to the green/red bounding box.	*'YOLOv8n identifies workers in real time. Notice the bounding box coordinates are mapped dynamically to the HTML5 Canvas.'*
Step 5: PPE Compliance Helmet & Vest Check	Highlight the right-side **Live Safety Status** panel showing per-worker compliance breakdown.	*'Our algorithm pairs detected equipment with worker bounding boxes via spatial IoU. It detects missing helmets or vests rather than simply counting gear.'*
Step 6: Obstacle Detection Hazard Persistence	Place a backpack, chair, or box inside the walkway zone. Note that it doesn't trigger immediately, but alerts after 10 frames with an orange hazard box.	*'This is our core Phase 1 feature. It computes Shapely polygon intersection. Notice temporal persistence: single-frame noise is ignored, eliminating false alarms.'*
Step 7: Incident Log Event Verification	Navigate to **Events (`/events`)**. Show the newly logged incident with timestamp, severity badge, and JPEG snapshot.	*'Every safety breach is logged in SQLite with an evidence snapshot for automated compliance auditing and future reporting.'*

5. Verification Testing & Troubleshooting

Automated Test Suite Verification (`pytest`):

Run the complete unit test suite covering YOLO engine loading, PPE compliance logic, Shapely zone overlap math, and temporal tracking:

```
python -m pytest -v tests/ # Expected: 10 passed in ~8 seconds
```

Frontend Production Compilation Check:

```
cd frontend && npm run build # Verifies TypeScript typing and builds production bundle
```

Common Troubleshooting Solutions:

- **PyTorch 2.6+ Checkpoint Warning:** Our codebase includes a pre-configured compatibility patch in engine.py and download_model.py allowing Ultralytics weights to load smoothly under PyTorch 2.6.
- **Port Already in Use:** If port 8000 or 3000 is occupied, run netstat -ano | findstr :8000 and terminate the PID, or launch Uvicorn with --port 8080.

- **Webcam Not Opening:** Verify camera permissions in Windows Settings $\rightarrow$ Privacy $\rightarrow$ Camera. If using an external USB webcam, change the source index from 0 to 1 in MonitoringPage.tsx.